

HDOC DAY 40

PROBLEM

Write a menu-driven C program that reads a student's first name, middle name, and last name. Use separate user-defined functions for each operation. Use call by reference by passing pointers wherever a function modifies a string or returns multiple results. Perform the following tasks:

1. Combine the first, middle, and last names to form and display the student's full name. Hint: Use strcpy() and strcat().
 - void combineNames(const char firstName[], const char middleName[], const char lastName[], char fullName[]);
2. Find and display the length of the full name. Hint: Use strlen().
 - void findLength(const char fullName[], size_t length);
3. Copy the full name into another string and display the copied string. Hint: Use strcpy().
 - void copyName(const char source[], char destination[]);
4. Compare the first name and last name and display whether they are equal or different. Hint: Use strcmp().
 - void compareNames(const char firstName[], const char lastName[], int comparisonResult);
5. Display the full name in reverse order, that is, last name, middle name, and first name. Hint: Use strcpy(), strcat(), and strtok() wherever appropriate.
 - void reverseNameOrder(const char firstName[], const char middleName[], const char lastName[], char reverseName[]);
6. Count and display the number of vowels, consonants, digits, and spaces in the full name. Hint: Use isalpha(), isdigit(), isspace(), tolower(), and strlen().
 - void countCharacters(const char fullName[], int vowels, int consonants, int digits, int spaces);
7. Read a word and check whether it occurs in the full name. Hint: Use strstr().
 - void searchWord(const char fullName[], const char word[], int searchResult);
8. Exit

CODE

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


#define MAX 100


/* Function prototypes */

void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[]);

void findLength(const char fullName[], size_t *length);

void copyName(const char source[], char destination[]);

void compareNames(const char firstName[], const char lastName[],
                 int *comparisonResult);

void reverseNameOrder(const char firstName[], const char middleName[],
                    const char lastName[], char reverseName[]);

void countCharacters(const char fullName[], int *vowels,
                   int *consonants, int *digits, int *spaces);

void searchWord(const char fullName[], const char word[], int *searchResult);
```

/* 1. Combine names */

```
void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[])
{
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}
```

/* 2. Find length */

```
void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}
```

/* 3. Copy name */

```
void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}
```

/* 4. Compare first and last names */

```
void compareNames(const char firstName[], const char lastName[],
```

```
        int *comparisonResult)
{
    *comparisonResult = strcmp(firstName, lastName);
}
```

/* 5. Reverse order of names */

```
void reverseNameOrder(const char firstName[], const char middleName[],
                     const char lastName[], char reverseName[])
{
    char temp[MAX];

    strcpy(reverseName, lastName);
    strcat(reverseName, " ");

    strcpy(temp, middleName);
    strcat(reverseName, temp);
    strcat(reverseName, " ");

    strcat(reverseName, firstName);
}
```

/* 6. Count vowels, consonants, digits and spaces */

```
void countCharacters(const char fullName[], int *vowels,
                    int *consonants, int *digits, int *spaces)
{
    size_t i;
```

```
char ch;
```

```
*vowels = 0;
```

```
*consonants = 0;
```

```
*digits = 0;
```

```
*spaces = 0;
```

```
for (i = 0; i < strlen(fullName); i++)
```

```
{
```

```
    ch = fullName[i];
```

```
    if (isalpha((unsigned char)ch))
```

```
    {
```

```
        ch = tolower((unsigned char)ch);
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' ||
```

```
            ch == 'o' || ch == 'u')
```

```
        {
```

```
            (*vowels)++;
```

```
        }
```

```
    else
```

```
    {
```

```
        (*consonants)++;
```

```
    }
```

```
}
```

```
else if (isdigit((unsigned char)ch))
```

```
{
```

```
    (*digits)++;
```

```

    }
    else if (isspace((unsigned char)ch))
    {
        (*spaces)++;
    }
}
}

```

/* 7. Search for a word */

```

void searchWord(const char fullName[], const char word[], int *searchResult)
{
    if (strstr(fullName, word) != NULL)
        *searchResult = 1;
    else
        *searchResult = 0;
}

```

/* Main function */

```

int main()
{
    char firstName[MAX], middleName[MAX], lastName[MAX];
    char fullName[MAX * 3];
    char copiedName[MAX * 3];
    char reverseName[MAX * 3];
    char word[MAX];

```

```
size_t length;

int comparisonResult;

int vowels, consonants, digits, spaces;

int searchResult;

int choice;


/* Read names */

printf("Enter first name: ");
scanf("%99s", firstName);


printf("Enter middle name: ");
scanf("%99s", middleName);


printf("Enter last name: ");
scanf("%99s", lastName);


/* Create full name initially */
combineNames(firstName, middleName, lastName, fullName);


do
{
    printf("\n===== MENU =====\n");
    printf("1. Display full name\n");
    printf("2. Find length of full name\n");
    printf("3. Copy full name\n");
    printf("4. Compare first and last name\n");
    printf("5. Display name in reverse order\n");
    printf("6. Count vowels, consonants, digits and spaces\n");
```

```
printf("7. Search for a word in full name\n");

printf("8. Exit\n");

printf("=====\n");

printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)
{
    case 1:
        combineNames(firstName, middleName, lastName, fullName);
        printf("Full name: %s\n", fullName);
        break;

    case 2:
        findLength(fullName, &length);
        printf("Length of full name: %zu\n", length);
        break;

    case 3:
        copyName(fullName, copiedName);
        printf("Copied name: %s\n", copiedName);
        break;

    case 4:
        compareNames(firstName, lastName, &comparisonResult);

        if (comparisonResult == 0)
```

```
    printf("First name and last name are equal.\n");  
else  
    printf("First name and last name are different.\n");  
break;
```

case 5:

```
reverseNameOrder(firstName, middleName,  
                  lastName, reverseName);  
printf("Name in reverse order: %s\n", reverseName);  
break;
```

case 6:

```
countCharacters(fullName, &vowels, &consonants,  
                &digits, &spaces);  
  
printf("Vowels   : %d\n", vowels);  
printf("Consonants : %d\n", consonants);  
printf("Digits   : %d\n", digits);  
printf("Spaces   : %d\n", spaces);  
break;
```

case 7:

```
printf("Enter word to search: ");  
scanf("%99s", word);  
  
searchWord(fullName, word, &searchResult);  
  
if (searchResult == 1)
```

```
        printf("\'%s\' occurs in the full name.\n", word);

    else

        printf("\'%s\' does not occur in the full name.\n", word);

    break;


case 8:

    printf("Exiting program...\n");

    break;


default:

    printf("Invalid choice! Please try again.\n");

}


} while (choice != 8);


return 0;

}
```